

TABLE II
Overview of Adhesion Tests

Fingernail test (peel test)	- Simple method; available at any time; preliminary test performed with coatings that are not too hard. The coating film is damaged using a thumbnail, followed by trying to scrape off the coating film with the thumbnail. Remember that both the hardness and the adhesion are assessed. - Significance of results is relatively rough as dependent on user's nail hardness. Suitable for pre-assessment if coating result shows signs of insufficient coating adhesion, e.g. orange peel formation or island formation. - As an alternative for hard coatings, a knife blade may be used (peel test).
Cross-cut test	- The test is standardised according to DIN EN ISO 2409. Within this method, a lattice is cut using the single-edge or multi-cross method, depending on the layer thickness of the coating applied, followed by an evaluation of the damaged area as a percentage of the lattice surface - Alternatively, at an international level, an evaluation in accordance with IPC-TM-650, 2.4.21E is also accepted or prescribed.
Pencil hardness test	Scratch method using a needle, described and assessed in accordance with the hardness for pencils, e.g. H6. Not particularly suitable for evaluating the adhesion of protective coatings since it is a hardness test.
Tear-off test with adhesive tape (tape test)	In this test, a strip of adhesive tape with a defined adhesive force (adhesive strength) is affixed to the coated surface and then stripped from the surface at a defined angle and speed. Along with the cross-cut test, this method is employed more and more often to assess the adhesion. Not suitable for silicone coatings. The evaluation parameters are defined by in-house regulations or by customer requirements.

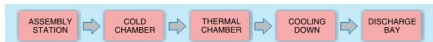


Fig. 3: Description of a burn-in test run

for this test must be chosen in accordance with common practice—in this example, 80°C). If the weight no longer drops, the assembly is stored at ambient temperature for two hours and then weighed again. On the basis of this data, the necessary drying time can also be determined.

Functional tests

Following the application of the protective coating, the coated item is usually either further processed or, in case of contract coating, dispatched to the customer. In the scope of quality control, fault reduction and monitoring of the process chain, a so-called functional test is performed after protective coating. Once the tests described above are completed, an electrical test or a test combining function and climatic loading is carried out. Quite often, a mechanical test is also included, in order to simulate the subsequent process of fitting into housings. In addition, the functionality of contact points such as plugs, test points and relays is verified. In



Fig. 4: Cold-heat tunnel system for detecting hidden defects in electronic components for the automotive industry (Source: Weiss Umwelttechnik GmbH)

view of the intended application, the functional test is often combined with a climatic and/or thermal loading. For this purpose, the relevant conditions of temperature and moisture are adjusted in a climatic chamber before the functionality of the assembly is checked. Burn-in test is a quality control tool used to check either complete devices or individual components of devices. Experience has shown that defective devices not identified in the manufacturing process will fail after a short time of continuous operation

and/or when operated under extreme conditions.

In order to detect hidden defects, a burn-in with thermal cycling is the suitable method to reduce early failures in the bathtub curve. As described in coating qualification, the stress factors are:

1. Temperature range (temperature lift, e.g. -40°C/+85°C)
2. Thermal cycling speed
3. Dwell time at peak temperatures
4. Number of temperature cycles

A burn-in may be realised as an individual station or as a tunnel system. For the two temperature peaks, and may be also in the dewed heat-up phase, functional tests are performed. To ensure traceability of the burn-in, the following points should be recorded:

1. Identification and assignment of the test specimen
2. Test temperature and dwell times
3. Results of functional tests
4. Evaluation of the test specimen (whether 'ok' or 'not ok')

Such climatic tests are often made as a serial arrangement where they are combined with the electrical function test as shown in Fig. 3. Also, primary stress factors related to moisture load are evaluated. If there is a risk of dewing, an alternating climatic load should be preferred over a constant climate load. In case of an alternating climate load, it must be ensured that dewing actually takes place as required, by using a dewing sensor for verification in case of doubt. These tests may also be performed in the scope of controlling the coating quality by random samples.

Climatic loads may be chosen as follows:

1. Thermal cycling: -40°C/+85°C, 6 cycles with 3-7K acceleration
2. Climate: 40°C/93% R.H., 6 cycles
3. Thermal shock: -40°C/+120°C, 24 cycles

This article is an extract from the book titled 'Conformal Coatings for Electronics Applications' by the author. The sources of information are mentioned in the book